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Session : Presentation on Technical Paper

Leveraging the Power of AI in Identifying Power Theft in Non Smart Meters

Presented By

Suhaas Mani, Manager, TP Western Odisha Distribution Limited

INTRODUCTION

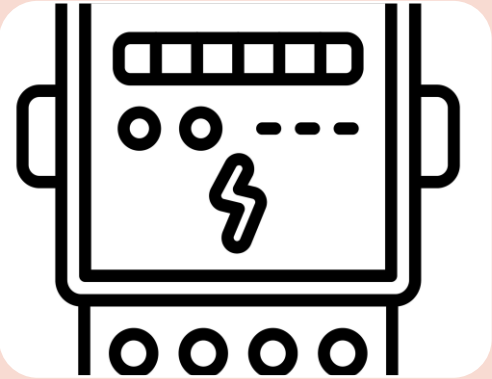
- Electricity theft is major contributor to AT&C losses
- Non-smart meters lack Communication, hence data analysis not possible to identify theft
- Door to Door visits are reactive, resource intensive
- Need low-cost transitional solution

Parameter	Value	Source of Information
AT%C loss	15% (As per MOP)	MOP
Total Consumer Base	34 Cr	India Climate and Energy Dashboard
No. of Smart Meters	5.28 Cr	https://www.energetica-india.net/news/indias-smart-meter-deployment-touches-528-crore-under-rdss?
Approximate Duration in which all Non-Smart Meters Can be replaced through smart meter	10 years	Assuming 2.8 Cr Smart Meter Installation Per Year

CONTEXT

Industry Challenge

1. Revenue Leakage in DISCOMs
2. Electricity theft in non-smart meter environment

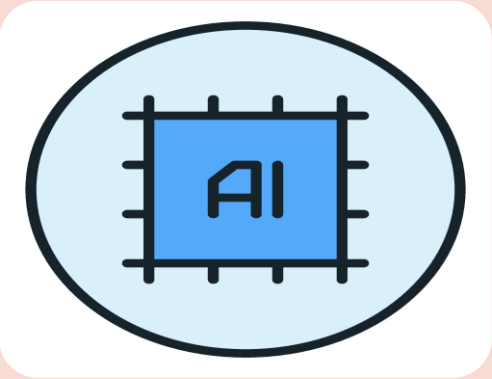


Limited Smart Meter

1. Majority consumers on conventional meters
2. No remote tamper communication

Symbols On Meter Display

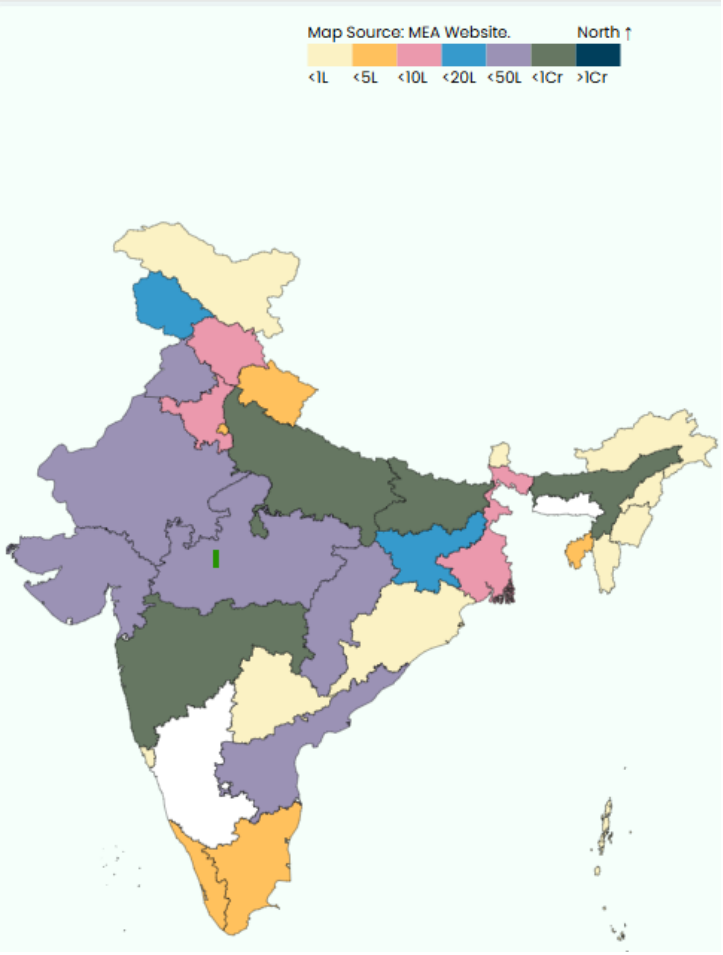
1. Visual Tamper Indicators Exist
2. Standardized symbols visible on meter display



Leveraging AI

1. Transform Images into Intelligence
2. AI-powered image analytics for theft detection

RELEVANCE



State Wise No. of Smart Meters Installed Till 15 Feb-26

01 Smart Meter Roll Out Ongoing

Transition phase — conventional meters still dominant.

02 Interim Revenue Protection

AI bridges enforcement gap

03 No Feeder Analytics Dependency

Independent of consumption pattern models

04 Scalable Across DISCOMs

Replicable using existing billing images

Transforming Theft Detection Through AI

Image analysis carried out for Already Identified Tampered Meters during door-to-door inspection

Already Detected Tampered Count	Tampered Symbol in Meter Display				Total Tampered Symbol	% Earth Symbol	% Meter Cover Open Symbol
	Earth Symbol	Meter Cover Open Symbol	Both Symbol	No Symbol			
5664	**3010**	**930**	**531**	**1193**	**4580**	**66%**	**20%**

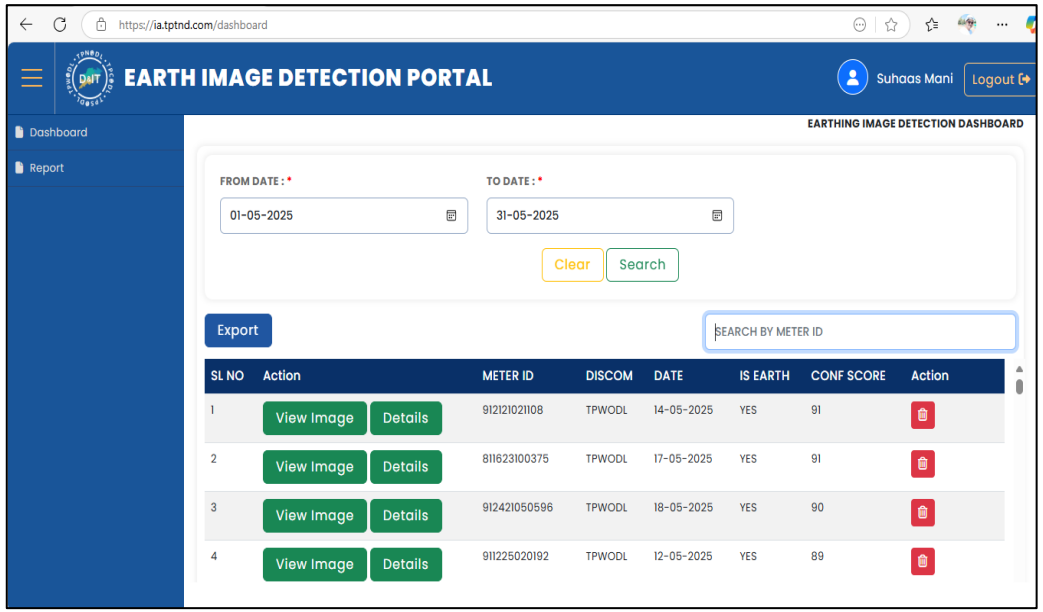
Why AI Image Detection is Now Strategic, Not Optional

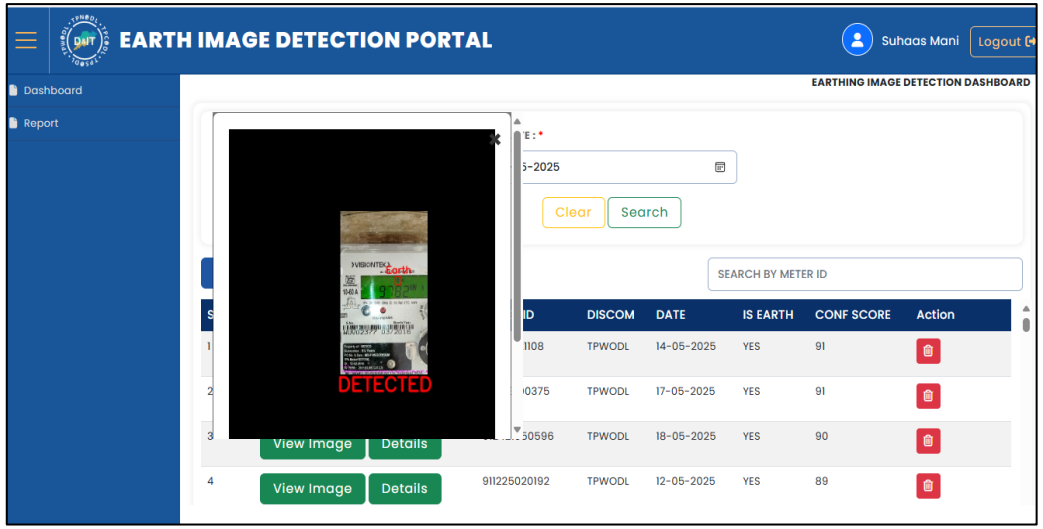
- 86% cases are **visually encoded in meter display**
- Depending on Meter make, tamper symbols follow **structured, repeatable visual patterns**
- AI (Computer Vision) can

1. Auto-detect Earth / Cover Open / Multiple tamper symbols
2. Enable bulk screening via field-captured meter photos
3. Instantly flag high-risk consumers for enforcement

Developing AI Model and Portal

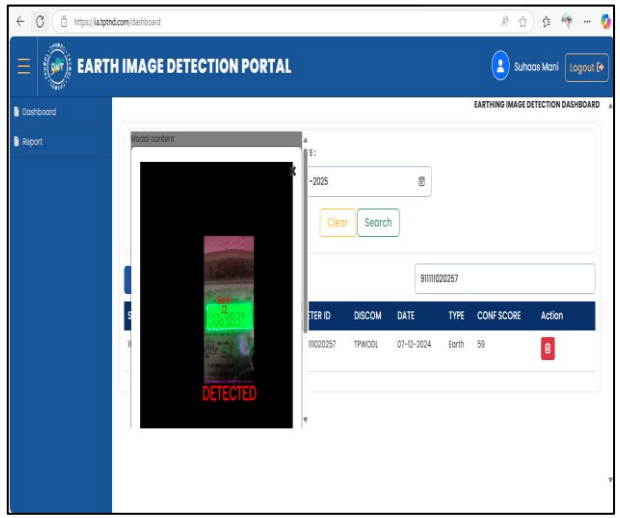
Sampling	Old Meters With Different Make and With Earth Symbol/ Cover Open Symbol is identified in Meter Testing Laboratory
Videography	High Quality Videos of Reading was Completed
Photographs	Thousands of photographs was captured
AI Model and Its Training	AI model is generated using Python and its library like OpenCV, NumPy etc
Portal to Fetch Sorted Cases	A portal is prepared where details of identified meters with earth symbol is sent.

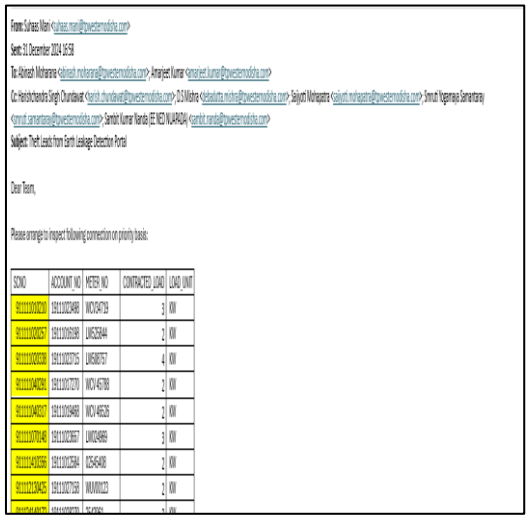


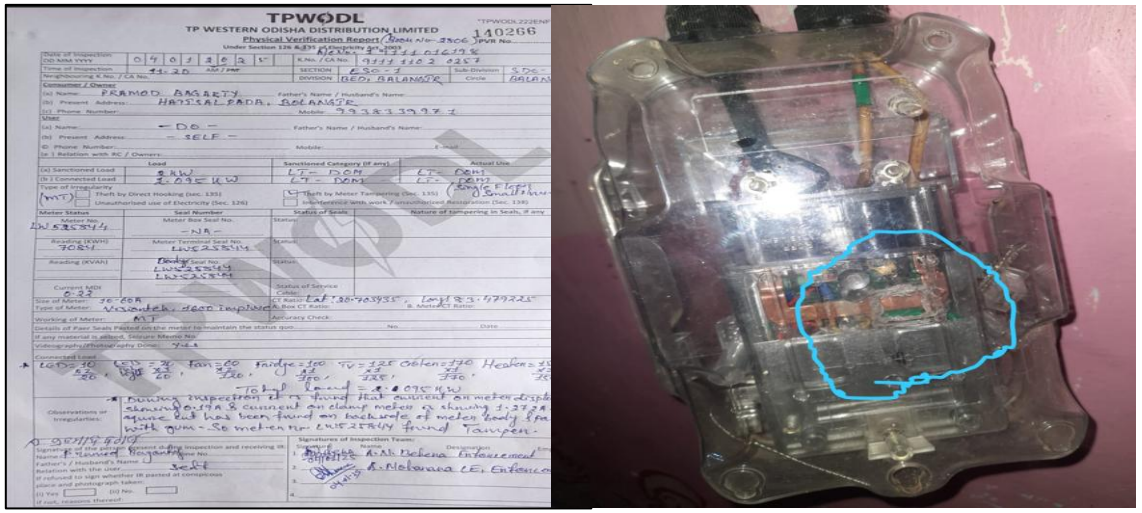


USE CASE / CASE STUDY

Parameters (From Jan-25 to Dec-25)	Count
Total Cases Flagged By System	4036
Correct Symbols Tagged	2029
Cases Referred	1220
Total Theft Booked	743
Hit Rate	61%
Amount Recovered from these cases	77.48 Lacs







Site Inspection Report:

- Square Cut Found on Meter Body & Pasted With Gum
- Meter Found Tampered

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THANK YOU

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Links/References (If any)